LABEL-FREE REACHABILITY IN JOINT-EMBEDDING PREDICTIVE ARCHITECTURES FOR SAFE REINFORCEMENT LEARNING

The Safety Dial: Irreversibility as a Deployment-Time Control

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Abstract

Safe reinforcement learning requires some specification of what counts as unsafe. Penalty-based methods require a cost function; constrained methods require a cost function and a threshold. A recent and rapidly developing line of work replaces both with *latent safety filters*, which use a learned world model to identify states from which failure is unavoidable and override the control policy before those states are reached. This removes the need to tune a penalty weight, but not the need for supervision: these methods are trained on trajectories annotated with visible failures, so the safety concept is still supplied by a human and the resulting filter is blind to any failure mode that was not labelled.

This proposal asks whether the failure set can instead be derived from dynamics alone. The proposed criterion is *irreversibility*: a transition is unsafe when the agent cannot return from it. Irreversibility is a property of the environment's dynamics rather than of a human's annotation, it requires no failure labels, and it is computable only with a world model, since it depends on counterfactual rollouts of actions that were not taken. The work develops return-reachability estimators in the latent space of a Joint-Embedding Predictive Architecture (JEPA) world model — a setting in which reachability analysis has not previously been attempted, existing latent safety filters being built on reconstructive world models — and evaluates them by zero-shot transfer: a supervised baseline and the proposed method are given identical data, the baseline additionally receives labels for one failure mode, and both are evaluated on a second, never-labelled failure mode. The resulting conservatism is then exposed as a runtime-adjustable parameter, the *Safety Dial*, allowing an operator to set caution at deployment rather than at training time. The central risk — that JEPA latents may not support a dynamically meaningful metric — is addressed by an explicit gating experiment scheduled before any method development.

Declaration

I, Sachin Mohan (Student Number: 2699183), hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of Bachelor of Science with Honours in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Contents

Chapter 1

Introduction

Reinforcement learning agents are increasingly deployed on physical systems, where unsafe behaviour has consequences that cannot be undone. The central practical question is not whether such agents should be safe but how safety is to be *specified*.

The dominant answer has been to specify it numerically. Reward shaping augments the task reward with a penalty term, $R'(s,a) = R(s,a) - \lambda C(s,a)$, and inherits a well-documented fragility: too small a λ and the agent ignores the penalty, too large and it becomes paralysed, with the correct value depending on quantities the designer rarely knows in advance (Skalse et al., 2022; Fu et al., 2025). Constrained formulations (Achiam et al., 2017) replace the weight with an explicit constraint, but require the designer to supply both a cost function C and a threshold, which relocates the specification burden rather than removing it (García and Fernández, 2015). ROSARL (Nangue Tasse et al., 2024) makes real progress here by deriving a provably sufficient penalty from intrinsic environment quantities, but still commits to a single scalar objective fixed before deployment.

A different answer has emerged from the model-based direction. *Latent safety filters* (Nakamura et al., 2025) generalise Hamilton–Jacobi reachability (Bansal et al., 2017) to the latent space of a learned world model, identifying states from which the agent is doomed to fail regardless of subsequent action and overriding the base policy before such states are entered. Recent extensions add out-of-distribution awareness (Seo et al., 2025) and runtime-adjustable constraints (AnySafe, 2026). These methods do not require a penalty weight, which is a genuine advance.

They do, however, require *failure labels*. The failure set that seeds the reachability computation is learned as a classifier over annotated observations, so a human still decides what unsafe looks like, and the filter can only anticipate the failure modes present in that annotation. A filter trained to recognise a toppled object has no representation of a dropped one. In deployment settings where the expensive failures are precisely the ones nobody anticipated, this is a substantive limitation, and it is acknowledged within that literature: Seo et al. (2025) observe that a world model extrapolating beyond its data may hallucinate safe outcomes in unseen failure regions, producing optimistic and unreliable safety estimates.

1.1 Proposed Direction

This proposal takes the position that the failure set need not be labelled at all, because a large and important class of failures shares a purely dynamical signature: they are *irreversible*. An object resting on a table can be pushed anywhere and returned; an object that has fallen from the table cannot be recovered by any action sequence available to the agent. Spilling, toppling, dropping and breaking are all one-way transitions. Irreversibility is a property of the transition structure, it requires no annotation, and — crucially — it cannot be evaluated from logged data alone, since it concerns what *would* happen under actions the agent did not take. It therefore requires exactly the object that latent safety filters already assume: a world model.

The proposal develops this idea in the latent space of a Joint-Embedding Predictive Architecture (LeCun, 2022; Maes et al., 2026), which predicts future latent states rather than future pixels. This choice is not incidental. Existing latent safety filters are built on reconstructive world models, and whether reachability analysis remains valid in a representation explicitly trained to discard control-irrelevant detail is an open question with real content: a JEPA may equally discard failure-relevant detail. The proposal treats this as a hypothesis to be tested rather than an assumption, and schedules the test before any method development.

1.2 Contributions

The intended contributions are:

1. A **label-free failure criterion** for latent safety filtering, based on return-reachability rather than on a supervised failure classifier.
2. The **first reachability analysis in a JEPA latent space**, together with an empirical characterisation of whether such latents admit a dynamically meaningful metric.
3. A **zero-shot transfer evaluation** in which the proposed method and a supervised baseline are tested on a failure mode that only the baseline's labels exclude, isolating the value of removing supervision.
4. The **Safety Dial**: a runtime-adjustable conservatism parameter derived from the reachability margin, allowing the safety–performance operating point to be chosen at deployment rather than fixed during training.

Chapter 2

Background and Related Work

2.1 Reinforcement Learning and Safety

Reinforcement learning is formalised as a Markov Decision Process (S, A, P, R, γ) in which an agent learns a policy $\pi(a \mid s)$ maximising expected discounted return (Sutton and Barto, 2018). García and Fernández (2015) identify two principal families of safe RL: those modifying the reward with a penalty, and those imposing explicit constraints (Achiam et al., 2017). Both require a cost function C, whose specification is frequently as difficult as specifying the reward itself (Skalse et al., 2022). Empirical comparison is standardised by Safety-Gymnasium (Ji et al., 2023), though its hazards are predominantly recoverable — an agent may enter a hazard region, incur cost, and leave — which makes it a poor fit for the irreversibility-focused evaluation proposed here.

2.2 Latent Safety Filters

Hamilton–Jacobi reachability (Bansal et al., 2017) computes the *backward reachable tube* of a designated failure set: the set of states from which failure is inevitable under worst-case disturbance. A safety filter monitors a task policy and overrides it at the boundary of this set.

Nakamura et al. (2025) generalise this to raw visual observation by performing the reachability computation in the latent space of a learned world model, obtaining safety filters for high-dimensional systems where an analytic dynamics model is unavailable. Seo et al. (2025) extend the framework with uncertainty quantification, addressing the case in which the world model is queried outside its training distribution and returns optimistic predictions. AnySafe (2026) parameterise the constraint so that it may be specified at runtime rather than fixed at training time, aligning a notion of similarity to failure through conformal calibration.

The common structure of these methods is: labelled failure observations $\rightarrow$ latent failure classifier $\rightarrow$ reachability value function $\rightarrow$ binary override. The proposed work departs from it at the first step and retains the remainder.

2.3 Reversibility and Viability

The idea that irreversibility is the safety-relevant property has an independent lineage. Grinsztajn et al. (2021) learn a reversibility estimator without reward supervision by training a classifier to predict temporal precedence — given two states observed in a trajectory, which came first — and show that high-confidence asymmetry in this prediction identifies irreversible transitions. Because precedence labels are recoverable from unlabelled trajectory data at no annotation cost, this provides a practical route to the criterion proposed here. In control theory, the corresponding object is the *viability kernel* (Aubin, 1991): the largest subset of the state space from which the system can remain within a constraint set indefinitely, defined as a fixed point rather than seeded from a labelled failure set.

2.4 Joint-Embedding Predictive Architectures

World models (Ha and Schmidhuber, 2018) predict environment dynamics; generative variants (Hafner et al., 2023) reconstruct future observations, committing capacity to visual detail irrelevant to control. JEPAs (LeCun, 2022) instead predict future latents conditioned on actions, computing the prediction loss in latent space. Their central training difficulty is representation collapse, addressed in recent work by the Sketched Isotropic Gaussian Regulariser (Balestriero and LeCun, 2025). LeWM (Maes et al., 2026) applies this to action-conditioned world modelling, achieving stable end-to-end training from pixels with a single effective hyper-parameter, and demonstrates that JEPA latents support linear probes for physical quantities. DINO-WM (Zhou et al., 2025) offers an alternative substrate using frozen pre-trained encoders.

Two properties of this substrate matter here. First, SIGReg regularises the latent distribution toward isotropy, which is what makes distances and volumes in that space comparable across directions and therefore makes a reachability measure well posed. Second, and in tension with the first, the architecture is explicitly trained to discard detail not required for latent prediction, raising the possibility that failure-relevant structure is discarded with it.

2.5 Research Gap

Latent safety filtering supplies the reachability machinery but depends on labelled failures. Reversibility estimation supplies a label-free criterion but has not been combined with reachability analysis or with a world model of this class. JEPA world models supply a task-agnostic latent substrate with a principled geometry, but have never been used for reachability analysis. The contribution of this proposal lies at the intersection: a reachability-based safety filter whose failure set is derived from irreversibility rather than annotation, computed in a JEPA latent.

Chapter 3

Research Objectives

3.1 Research Question

Can the failure set required by a latent safety filter be derived from irreversibility in the dynamics — without failure labels — and does such a filter, computed in the latent space of a JEPA world model, detect failure modes that a supervised latent safety filter trained on different labelled failures cannot?

3.2 Research Goals

1. **Establish latent metric validity.** Determine empirically whether distance in a JEPA latent space corresponds to dynamical proximity — action-steps between states — rather than to visual similarity alone. This is a prerequisite for all subsequent work and is treated as a gating result.

2. **Develop label-free irreversibility estimators.** Construct and compare estimators of return-reachability in latent space, including a rollout-based return-distance measure and a precedence-classifier estimator in the style of Grinsztajn et al. (2021).

3. **Demonstrate zero-shot failure transfer.** Show that the resulting failure set generalises to failure modes absent from any label set, in direct comparison with a supervised latent safety filter.

4. **Expose conservatism as a runtime control.** Derive the Safety Dial from the reachability margin and characterise the resulting safety–performance curve.

3.3 Hypotheses

- **H1 (Metric validity).** Distance in the SIGReg-regularised JEPA latent correlates with action-steps between states, not merely with observational similarity.

- **H2 (Detectability).** A return-reachability score computed by counterfactual rollout separates irrecoverable from recoverable states substantially better than chance, and better than forward-dispersion measures, which are confounded by residual actuator freedom.

- **H3 (Transfer).** On a failure mode excluded from the supervised baseline's labels, the label-free method achieves materially higher detection rate than the supervised baseline, while remaining competitive on the labelled failure mode.

- **H4 (Dial monotonicity).** Varying the reachability threshold at deployment produces a monotone trade-off between irreversible-failure rate and task success, recovered from a single trained model rather than from repeated training runs.

Chapter 4

Research Methodology

The work is organised as five stages, gated so that the project terminates in a defensible outcome at each boundary.

4.1 Stage 0: Substrate and Environment

A released LeWM checkpoint (Maes et al., 2026) is used rather than a fresh reproduction; world-model training is infrastructure, not contribution. The evaluation environment must contain genuinely irreversible failures, which excludes Safety-Gymnasium hazards. OGBench-Cube (Park et al., 2025) is the primary candidate: a cube pushed from the table is unrecoverable by any available action, while a cube on the table is freely repositionable. A data-collection harness records (observation, action, latent, t, trajectory id, simulat with simulator state reserved exclusively for evaluation.

4.2 Stage 1: Latent Metric Validity (Gating)

Before any safety measure is constructed, the latent geometry is tested. State pairs are sampled and latent distance is regressed against (i) true simulator-state distance and (ii) the number of action steps separating the pair in observed trajectories. A latent that tracks only (i) through visual similarity, without tracking (ii), cannot support reachability analysis.

Because physical quantities are linearly decodable from JEPA latents (Maes et al., 2026), a complementary probe-space variant is constructed: a linear probe for object position is trained, and the reachability measure is computed within the probe subspace. This yields an interpretable measure in which the contribution of each physical quantity is directly visible, at the cost of requiring position supervision. Note that this is supervision of a generic, task-agnostic quantity, not of the safety concept itself; the method remains free of *failure* labels, and the distinction is stated explicitly in all reporting. The raw-latent and probe-space measures are computed in parallel throughout, with agreement between them serving as empirical validation of the raw latent metric.

4.3 Stage 2: Irreversibility Estimation

4.3.1 Return-reachability by rollout

For a candidate transition from z_t to z_{t+k}, K action sequences are sampled from a distribution held *fixed across all evaluated states*, rolled forward through the JEPA predictor from z_{t+k}, and the score is taken as the minimum distance from any resulting latent back to z_t:

$$\rho(z_t \to z_{t+k}) \;=\; \min_{j\in\{1,\ldots,K\}} d\Big(\hat{z}^{(j)}_{t+k+H},\, z_t\Big)\,.$$

A large ρ indicates that no sampled action sequence returns the system to its prior state, i.e. the transition was one-way. Holding the action distribution fixed ensures that differences in ρ reflect dynamics rather than differences in available control authority.

Forward dispersion — the spread of terminal latents, an empowerment-style measure — is implemented as a comparison, with the expectation that it underperforms: when a cube falls from the table, the arm retains full freedom, so aggregate dispersion remains high even though the state has become irrecoverable. Where a covariance-based statistic is required, effective rank $(\sum_i \lambda_i)^2/\sum_i \lambda_i^2$ is used in preference to log-determinant, which is undefined when the sample covariance is rank-deficient.

4.3.2 Return-reachability by precedence

Following Grinsztajn et al. (2021), a classifier is trained on unlabelled trajectory data to predict, for a latent pair (z_i, z_j), whether i precedes j. Confident asymmetry indicates an irreversible transition. Precedence labels are free, and the resulting estimator requires no rollout at inference, which addresses the computational cost of the rollout-based measure.

4.3.3 Validation

Approximately one hundred states are annotated as recoverable or irrecoverable *for evaluation only*. To avoid selection bias toward visually obvious cases, these are generated programmatically — irrecoverable states follow a scripted push-off-edge manoeuvre, recoverable states are matched mid-manipulation states with comparable arm configurations — rather than chosen by inspection. Estimators are compared by AUROC against these annotations. Two controls are mandatory: scoring states at joint limits, at which control authority is reduced but nothing is broken, to confirm the measure does not track actuator freedom; and sweeping the rollout horizon H, to confirm the measure does not simply track accumulated predictor drift. In-distribution states with no failure provide the null.

4.4 Stage 3: Zero-Shot Failure Transfer

This stage produces the principal empirical claim. Two irreversible failure modes are defined in the environment (for instance, cube removed from the workspace, and cube wedged in an unrecoverable configuration). A supervised latent safety filter in the style of Seo et al. (2025) is trained with labels for mode A only. The proposed method is trained on the identical dataset without labels. Both are evaluated on mode B.

The critical control is that the shared unlabelled dataset must *contain* instances of mode B. Without this, the comparison measures data coverage rather than method, and the result is void. Dataset composition is verified and reported before the comparison is run.

4.5 Stage 4: The Safety Dial and Evaluation

The reachability score induces a margin, thresholded at a level d selected at deployment. Calibration is performed on held-out offline data so that d carries a distributional interpretation rather than being an arbitrary constant; conformal prediction supplies the machinery for the associated coverage statement. Evaluation reports the irreversible-failure rate and task success as d is varied, and compares the compute required to recover the full trade-off curve against penalty-swept baselines, which require one training run per operating point.

Where time permits, a value function conditioned on d and learned over frozen latents is investigated as a terminal cost, extending the safety guarantee beyond the planning horizon.

Chapter 5

Research Plan

5.1 Time Plan

Stage	Period	Activities
Stage 0	Late August	Literature consolidation; LeWM checkpoint running; environment selection; data-collection harness.
Stage 1	Early September	Latent metric validity experiments; probe training; **gating decision**.
Stage 2	September	Rollout and precedence estimators; annotation harness; AUROC evaluation with controls.
Stage 3	October	Supervised baseline reproduction; zero-shot transfer comparison; dataset composition verification.
Stage 4	Late October	Safety Dial calibration; trade-off curves; compute comparison; ablations.
Write-up	November	Analysis, additional experiments indicated by results, thesis writing, supervisor revisions, submission.

Table 5.1: Project timeline (August 2026 – November 2026).

5.2 Success Criteria

In decreasing order of necessity:

(a) A characterisation of whether JEPA latent distance carries dynamical meaning, with the probe-space comparison. This is a reportable result whether the answer is positive or negative.

(b) A label-free irreversibility estimator achieving AUROC ≥ 0.8 against held-out annotations, with both confounds controlled.

(c) Demonstrated zero-shot detection of an unlabelled failure mode exceeding a supervised baseline on the same data.

(d) A calibrated Safety Dial producing a monotone safety–performance trade-off from a single trained model.

5.3 Risks and Mitigations

The latent metric fails Stage 1. If distance in the JEPA latent tracks appearance rather than dynamics, the approach cannot proceed as designed. Mitigation: the probe-space formulation provides a fallback in which the measure is computed over decoded physical coordinates. The negative result is itself publishable and bears directly on whether non-reconstructive world models can support safety analysis at all.

Irreversibility does not coincide with danger. A correctly placed object has few available futures and is entirely safe; conversely, some hazards are recoverable. Mitigation: the claim is scoped explicitly to irreversible failures — the subset where labels are most expensive and consequences most severe — rather than to safety in general. This is stated as a limitation, not defended against.

The supervised baseline is not reproducible in time. Reimplementing a latent safety filter is a substantial engineering task. Mitigation: begin with the authors' released code where available; failing that, an ablated version of the proposed method using labelled failures as its boundary condition serves as an internal supervised control.

Computational cost of nested rollouts. Return-reachability requires a second level of rollout per candidate transition. Mitigation: the precedence estimator requires no rollout at inference and is developed in parallel for exactly this reason; pre-computation and reduced K are available as further reductions.

Concurrent work. This is an active area with a high preprint rate. Mitigation: the specific combination — label-free failure sets from irreversibility, in a non-reconstructive latent — is monitored continuously on arXiv and OpenReview, and the contribution is scoped so that the Stage 1 characterisation retains value independently.

Chapter 6

Conclusion

Existing latent safety filters have removed the penalty weight from safe reinforcement learning but retained the requirement for human-supplied failure labels, and are consequently blind to failure modes that were never annotated. This proposal argues that for an important class of failures — the irreversible ones — the label is unnecessary, because irreversibility is a dynamical property computable from a world model by counterfactual rollout.

The work develops return-reachability estimators in the latent space of a JEPA world model, a substrate in which reachability analysis has not previously been attempted and whose validity for this purpose is itself an open question tested at the outset. The principal empirical claim is established by zero-shot transfer to an unlabelled failure mode, isolating the value of removing supervision rather than merely matching a supervised method. The resulting conservatism is exposed as a deployment-time control, so that the safety–performance operating point is chosen by an operator with knowledge of the deployment context rather than by an engineer guessing in advance.

The project is deliberately gated. If JEPA latents prove not to support a dynamically meaningful metric, that finding is reported and the probe-space formulation preserves a coherent contribution. If they do, the path to the full result is direct.

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